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Selecting garbage collection processes

Abstract

A system executes a first virtual thread, including storing a first shared object in a first private memory region, storing a pointer in a shared memory region, and determining that the pointer includes a reference to a location of the first shared object in the first private memory region. Responsive to determining that the pointer includes the reference, the system designates the first virtual thread as disqualified from eligibility for execution of a sweeping thread-local garbage collection process and augments the pointer to indicate that the pointer includes the reference to the location of the first shared object. The system executes a second virtual thread, including loading the pointer and determining that the pointer has been augmented. Responsive to determining that the pointer has been augmented, the system designates the second virtual thread as disqualified from eligibility for execution of the sweeping thread-local garbage collection process.

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Background/Summary

TECHNICAL FIELD

(1) The present disclosure relates to garbage collection processes utilized by runtime environments to reclaim memory that is no longer in use or that is no longer accessible. More particularly, the present disclosure relates to selection of a garbage collection process from a group of garbage collection processes.

BACKGROUND

(2) A runtime environment may utilize a garbage collection process to reclaim memory that is no longer in use or that is no longer accessible to threads that may be executing in the runtime environment. A garbage collection process may be built into the runtime environment and may be executed automatically. By reclaiming memory that is no longer in use or that is no longer accessible, the reclaimed memory can be reallocated to other purposes. Additionally, reclaiming memory prevents memory leaks, whereby system memory is gradually consumed because of memory remaining unreleased even after the memory is no longer being utilized.

(3) The content of this background section should not be construed as prior art merely by virtue of its presence in this section.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The embodiments are illustrated by way of example and not by way of limitation in the figures of the accompanying drawings. It should be noted that references to “an” or “one” embodiment in

this disclosure are not necessarily to the same embodiment, and they mean at least one. In the drawings:

(2) FIG. 1 illustrates an example computing architecture in accordance with one or more embodiments;

(3) FIG. 2 is a block diagram illustrating one embodiment of a computer system suitable for implementing methods and features described herein;

(4) FIG. 3 illustrates an example virtual machine memory layout in block diagram form according to an embodiment;

(5) FIG. 4 illustrates an example frame in block diagram form according to an embodiment;

(6) FIGS. 5A-5C are flowcharts that illustrate example operations pertaining to selection of garbage collection processes; and

(7) FIG. 6 shows a block diagram that illustrates a computer system in accordance with one or more embodiments.

DETAILED DESCRIPTION

(8) In the following description, for the purposes of explanation, numerous specific details are set forth to provide a thorough understanding. One or more embodiments may be practiced without these specific details. Features described in one embodiment may be combined with features described in a different embodiment. In some examples, well-known structures and devices are described with reference to a block diagram form to avoid unnecessarily obscuring the present disclosure. 1. GENERAL OVERVIEW 2. DEFINITIONS 3. ARCHITECTURAL OVERVIEW 3.1 EXAMPLE CLASS FILE STRUCTURE 3.2 EXAMPLE VIRTUAL MACHINE ARCHITECTURE 3.3 LOADING, LINKING, AND INITIALIZING 4. EXAMPLE OPERATIONS 4.1 SELECTING GARBAGE COLLECTION PROCESSES 4.2 DISQUALIFYING THREADS FROM SWEEPING THREAD-LOCAL GARBAGE COLLECTION 5. HARDWARE OVERVIEW 6. MISCELLANEOUS; EXTENSIONS

1. General Overview

(9) A system selects a garbage collection (GC) process for a thread based on whether or not the thread has encountered a shared object that is (a) stored in a private memory region and (b) may be accessed by one or more additional threads. If the thread does not encounter any shared objects that are stored in a private memory region, the system may execute a sweeping thread-local garbage collection (sweeping TLGC) process that reclaims memory associated with the thread when or after the thread terminates. If the thread encounters a shared object that is stored in a private memory region and may be accessible by another thread, the system executes a selective garbage collection (selective GC) process that selectively reclaims a subset of memory blocks corresponding to a subset of private objects that are inaccessible from any thread. Any shared objects that are stored in a private memory region are not reclaimed in the selective GC process because such shared objects might be accessible by one or more additional threads. The selective GC process may include a thread-local selective GC process that includes identifying and reclaiming memory blocks corresponding to a subset of private objects from a private memory region allocated for the terminated thread. Additionally, or alternatively, the selective GC process may include a global selective GC process that includes identifying and reclaiming memory blocks corresponding to a subset of private objects from a plurality of private memory regions.

(10) In one example, execution of a first thread may include storing a first set of objects in a first private memory region. Upon or after termination of the first thread, the system may execute the sweeping TLGC process in response to determining that the first set of objects is exclusive of any shared objects accessible by one or more additional threads. Additionally, or alternatively, execution of a second thread may include storing a second set of objects in a second private memory region. The second set of objects may include at least one shared object that may be accessed by one or more additional threads. Upon or after termination of the second thread, the system may execute the selective GC process in response to determining that the second set of

objects includes at least one shared object accessible by one or more additional threads.

(11) In one example, execution of a first thread may include loading a first set of objects. Upon or after termination of the first thread, the system may execute the sweeping TLGC process to reclaim a first private memory region allocated for the first thread in response to determining that the first set of objects is exclusive of any shared objects stored in any private memory region and accessible by one or more additional threads. Additionally, or alternatively, execution of a second thread may include loading a second set of objects. The second set of objects may include at least one shared object that is stored in a private memory region and that may be accessed by one or more additional threads. Upon or after termination of the second thread, the system may execute the selective GC process in response to determining that the second set of objects includes at least one shared object that is stored in a private memory region and that may be accessed by one or more additional threads.

(12) In one example, the system may disqualify virtual threads from eligibility for execution of the sweeping TLGC process when the respective virtual thread stores or loads a pointer that includes a reference to a shared object in the private memory region. For virtual threads that are not disqualified, the sweeping TLGC process may be executed when the virtual thread exits. For virtual threads that are disqualified, the sweeping TLGC process is not executed. Instead, a selective GC process may be executed to selectively reclaim a subset of memory blocks corresponding to a subset of private objects that are inaccessible from any thread.

(13) In one example, a first virtual thread may store a pointer to a shared memory region, and the system may determine that the pointer includes a reference to a location of a shared object in a first private memory region. In response to determining that the pointer references the first private memory region, the system may designate the first virtual thread as disqualified from eligibility for execution of the sweeping TLGC process. Additionally, the system may augment the pointer to indicate that the pointer includes the reference to the first private memory region. The augmentation to the pointer may serve as an indicator for the system to disqualify other threads that load the pointer from eligibility for execution of the sweeping TLGC process. In one example, when a second virtual thread loads a pointer, the system may determine that the pointer has been augmented to indicate that the pointer includes a reference to a location of a shared object in a private memory region. In response to determining that the pointer has been augmented to indicate that the pointer references the private memory region, the system may designate the second virtual thread as disqualified from eligibility for execution of the sweeping TLGC process.

(14) In one example, the augmentation to the pointer may trigger an error in response to a virtual thread loading the pointer. The system may determine the error, and responsive to determining the error, the system may execute an error-handling operation on the pointer. The error-handling operation is configured to distinguish errors induced by the pointer having been augmented to indicate that the pointer includes a reference to a private memory region from other known or unknown errors that may arise. When the error-handling operation determines that the error is attributable to the pointer having been augmented to indicate that the pointer references a private memory region, the system designates the virtual thread that loaded the pointer as disqualified from eligibility for execution of the sweeping TLGC process. Additionally, the system may utilize the error-handling operation to resolve the error, including determining a location of the shared object referenced by the pointer. Upon having resolved the error, the virtual thread that loaded the pointer may proceed to access the shared object.

(15) By augmenting pointers that include a reference to a private memory region when the pointer is stored, the system may detect the augmentation when the pointer is loaded, and the system may disqualify the virtual thread that loaded the pointer from eligibility for execution of the sweeping TLGC process. Additionally, by utilizing an augmentation that triggers an error when the pointer is loaded, the system may detect the augmentation without having to directly interrogating each individual pointer that is loaded. As a result, the sweeping TLGC process may be executed for

virtual threads that are not disqualified without risk of inadvertently reclaiming memory blocks associated with a shared objects that might be accessed by other virtual threads. In one example, execution of a sweeping TLGC process may be particularly advantageous in the context of virtual threads due to the large number of virtual threads that may be managed by a runtime environment. However, a sweeping TLGC process may also be executed with respect to other types of threads, and the aforementioned advantages may be realized.

(16) One or more embodiments described in this Specification and/or recited in the claims may not be included in this General Overview section.

2. Definitions

(17) As used herein, the term “garbage collection” or “GC” refers to a process of reclaiming memory that is no longer in use.

(18) As used herein, the term “sweeping thread-local garbage collection” or “sweeping TLGC” refers to a process of reclaiming an entirety of a private memory region associated with a particular thread. A sweeping TLGC process may be executed with respect to one or more of the following: a virtual thread, a carrier thread, a platform thread, or an operating system thread that is managed within a runtime environment by a platform thread.

(19) As used herein, the term “selective garbage collection” or “selective GC” refers to a process of selectively reclaiming from at least one private memory region, a subset of memory blocks allocated for a subset of private objects that are inaccessible from any thread. The process of selectively reclaiming the subset of memory blocks may include identifying memory blocks that are inaccessible from any thread and reclaiming the identified memory blocks. A selective GC process may be executed with respect to one or more of the following: a virtual thread, a carrier thread, a platform thread, or an operating system thread that is managed within a runtime environment by a platform thread.

(20) As used herein, the term “global selective garbage collection” or “global selective GC” refers to a selective GC process that is executed with respect to a plurality of threads. A global selective GC process may be executed with respect to one or more of the following: a plurality of virtual threads, a plurality of carrier threads, a plurality of platform threads, or a plurality of operating system threads that are respectively managed within a runtime environment by a platform thread.

(21) As used herein, the term “thread-local selective garbage collection” or “thread-local selective GC” refers to a selective GC process that is executed with respect to a particular thread. A thread-local selective GC process may be executed with respect to one or more of the following: a virtual thread, a carrier thread, a platform thread, or an operating system thread that is managed within a runtime environment by a platform thread.

(22) As used herein, the term “thread” refers to an independent set of one or more instructions representing a unit of execution within a process executable by a computing system. In one example, a thread may be the smallest unit of processing that can be scheduled. Multiple threads may execute concurrently and/or independently of one another. The term “thread” includes threads that are managed by an operating system as well as threads that are managed by a runtime environment. Threads that are managed by an operating system may include operating system threads. Threads that are managed by a runtime environment may include platform threads and virtual threads.

(23) As used herein, the term “operating system thread” refers to a thread that is managed by an operating system kernel at an operating system-level, and that is managed within a runtime environment by a platform thread. The runtime environment provides a layer of abstraction, and the platform thread may manage the execution and behavior of the underlying operating system thread within the constraints and abstractions provided by the runtime environment.

(24) As used herein, the term “platform thread” refers to a thread that is implemented by a runtime environment as a wrapper around an operating system thread and that is schedule for execution by the operating system. The runtime environment may map a particular platform thread to a

particular operating system thread in a one-to-one relationship. The operating system thread remains occupied by the platform thread until the platform thread is terminated. In one example, a platform thread may execute Java code on an underlying operating system thread. In one example, a platform thread may be an instance of the java.lang. Thread class that is implemented as a wrapper around an operating system thread. An operating system thread can be managed within a runtime environment by a platform thread also executing in the runtime environment.

(25) As used herein, the term “virtual thread” refers to a thread that is implemented by a runtime environment and that is scheduled by the runtime environment for execution via a carrier thread such as a platform thread. The runtime environment may schedule multiple virtual threads for execution across a group of carrier threads. Typically, many virtual threads are scheduled across a small number of carrier threads. For a particular virtual thread, the runtime environment may select a carrier thread from the group of carrier threads and may mount the virtual thread to the selected carrier thread. The operating system may schedule the carrier thread for execution, thereby allowing the virtual thread to execute instructions via the operating system thread that is mapped to the carrier thread. Subsequently, the runtime environment may unmount a virtual thread from a carrier thread, thereby making the carrier thread available to another virtual thread. The runtime environment may schedule the multiple virtual threads for execution across the group of carrier threads to provide improved throughput relative to executing operations directly via platform threads. In one example, when a virtual thread encounters a blocking operation, such as a blocking I/O operation, the runtime environment may execute a non-blocking operating system call for execution via the operating system thread mapped to the carrier thread. Additionally, the runtime environment may suspend execution of the virtual thread and unmount the virtual thread from the carrier thread, so the carrier may be utilized by another virtual thread while the blocking operation is executed. The operating system may execute the operation asynchronously without blocking execution of the carrier thread or the operating system thread. When the blocking operation is ready to complete, such as when bytes have been received on a socket, the runtime environment may schedule the virtual thread to be mounted on another carrier thread to resume execution of the virtual thread. A virtual thread may be scheduled to execute via one or more carrier threads over the lifecycle of the virtual thread.

(26) As used herein, the term “carrier thread” refers to a thread that a runtime environment schedules to execute one or more operations of a virtual thread via an operating system thread. The carrier thread may serve as an intermediary between the virtual thread, scheduled by the runtime environment, and an operating system thread, scheduled by the operating system. The use of carrier threads allows the runtime environment to manage a larger number of virtual threads without the overhead that would be associated with creating and managing a one-to-one mapping between virtual threads and individual operating system threads.

(27) As used herein, the term “blocking operation” refers to an operation of a thread that halts the progress of further code execution of the thread until the operation is complete.

(28) As used herein, the term “blocking I/O operation” refers to a blocking operation that involves an interaction with an external resource, such as reading from or writing to files, making network requests, or accessing peripheral devices. A blocking I/O operation may include an input operation, such as a read operation, or an output operation, such as a write operation.

3. Architectural Overview

(29) FIG. 1 illustrates an example architecture according to at least one embodiment. Software and/or hardware components described with relation to the example architecture may be omitted or associated with a different set of functionalities than described herein. Software and/or hardware components, not described herein, may be used within an environment in accordance with one or more embodiments. Accordingly, the example environment should not be constructed as limiting the scope of any of the claims.

(30) As illustrated in FIG. 1, a computing architecture **100** includes source code files **101** that are

compiled by a compiler **102** into class files **103** representing the program to be executed. The class files **103** are then loaded and executed by an execution platform **112**. The execution platform **112** includes a runtime environment **113**, an operating system **111**, and one or more application programming interfaces (APIs) **110** that enable communication between the runtime environment **113** and the operating system **111**. The runtime environment **113** includes a virtual machine **104** that includes various components, such as a memory manager **105**, a class file verifier **106**, a class loader **107**, an interpreter **108**, and a just-in-time (JIT) compiler **109**. The memory manager **105** may include a garbage collector. The garbage collector may execute garbage collection operations. The class file verifier **106** may check the validity of class files **103**. The class loader **107** may locate and build in-memory representations of classes. The interpreter **108** may execute the virtual machine **104** code. The JIT compiler **109** may producing optimized machine-level code.

(31) In an embodiment, the computing architecture **100** includes source code files **101** that contain code that has been written in a particular programming language, such as Java, C, C++, C#, Ruby, Perl, and so forth. Thus, the source code files **101** adhere to a particular set of syntactic and/or semantic rules for the associated language. For example, code written in Java adheres to the Java Language Specification. However, since specifications are updated and revised over time, the source code files **101** may be associated with a version number indicating the revision of the specification. The exact programming language used to write the source code files **101**. Various programming languages may be utilized for various different embodiments.

(32) In various embodiments, the compiler **102** may convert the source code, written according to a specification directed to the convenience of the programmer, to either machine or object code that is executable directly by the particular machine environment. Additionally, or alternatively, the compiler **102** may convert the source code to an intermediate representation (“virtual machine code/instructions”) such as bytecode that is executable by a virtual machine **104**. The virtual machine **104** may be capable of running on top of a variety of particular machine environments. The virtual machine instructions are executable by the virtual machine **104** in a more direct and efficient manner than the source code. Converting source code to virtual machine instructions includes mapping source code functionality from the language to virtual machine functionality that utilizes underlying resources, such as data structures. Often, functionality that is presented in simple terms via source code by the programmer is converted into more complex steps that map more directly to the instruction set supported by the underlying hardware where the virtual machine **104** resides.

(33) In general, programs are executed either as a compiled or an interpreted program. When a program is compiled, the code is transformed globally from a first language to a second language before execution. Since the work of transforming the code is performed ahead of time; compiled code tends to have excellent run-time performance. In addition, since the transformation occurs globally before execution, the code can be analyzed and optimized using techniques such as constant folding, dead code elimination, inlining, and so forth. However, depending on the program being executed, the startup time can be significant. In addition, inserting new code would require the program to be taken offline, re-compiled, and re-executed. For many dynamic languages (such as Java) that are designed to allow code to be inserted during the program's execution, a purely compiled approach may be inappropriate. When a program is interpreted, the code of the program is read line-by-line and converted to machine-level instructions while the program is executing. As a result, the program has a short startup time (can begin executing almost immediately), but the run-time performance is diminished by performing the transformation on the fly. Furthermore, since each instruction is analyzed individually, many optimizations that rely on a more global analysis of the program cannot be performed.

(34) In some embodiments, the virtual machine **104** includes an interpreter **108** and a JIT compiler **109** (or a component implementing aspects of both), and executes programs using a combination of interpreted and compiled techniques. For example, the virtual machine **104** may initially begin by

interpreting the virtual machine instructions representing the program via the interpreter **108** while tracking statistics related to program behavior, such as how often different sections or blocks of code are executed by the virtual machine **104**. Once a block of code surpasses a threshold (is “hot”), the virtual machine **104** invokes the JIT compiler **109** to perform an analysis of the block and generate optimized machine-level instructions that replaces the “hot” block of code for future executions. Since programs tend to spend a significant proportion of time executing a small portion of overall code, compiling just the “hot” portions of the program can provide similar performance to fully compiled code, but without the start-up penalty. Furthermore, although the optimization analysis is constrained to the “hot” block being replaced, there still exists far greater optimization potential than converting each instruction individually. There are a number of variations on the above-described example, such as tiered compiling.

(35) To provide clear examples, the source code files **101** have been illustrated as the “top level” representation of the program to be executed by the execution platform **112**. Although the computing architecture **100** depicts the source code files **101** as a “top level” program representation, in other embodiments the source code files **101** may be an intermediate representation received via a “higher level” compiler that processed code files in a different language into the language of the source code files **101**. Some examples in the following disclosure assume that the source code files **101** adhere to a class-based object-oriented programming language. However, this is not a requirement to utilizing the features described herein.

(36) In an embodiment, compiler **102** receives as input the source code files **101** and converts the source code files **101** into class files **103** that are in a format expected by the virtual machine **104**. For example, in the context of the JVM, the Java Virtual Machine Specification defines a particular class file format, and the class files **103** are expected to adhere to the particular class file format. In some embodiments, the class files **103** contain the virtual machine instructions that have been converted from the source code files **101**. However, in other embodiments, the class files **103** may contain other structures as well, such as tables identifying constant values and/or metadata related to various structures (classes, fields, methods, and so forth).

(37) The following discussion assumes that each of the class files **103** represents a respective “class” defined in the source code files **101** (or dynamically generated by the compiler **102**/virtual machine **104**). However, the aforementioned assumption is not a strict requirement and will depend on the implementation of the virtual machine **104**. Thus, the techniques described herein may still be performed regardless of the exact format of the class files **103**. In some embodiments, the class files **103** are divided into one or more “libraries” or “packages” that respectively include a collection of classes that provide related functionality. For example, a library may contain one or more class files that implement input/output (I/O) operations, mathematics tools, cryptographic techniques, graphics utilities, and so forth. Further, some classes (or fields/methods within those classes) may include access restrictions that limit their use to within a particular class/library/package or to classes with appropriate permissions.

3.1 Example Class File Structure

(38) FIG. 2 illustrates an example structure for a class file **200** in block diagram form according to an embodiment. To provide clear examples, the remainder of the disclosure assumes that the class files **103** of the computing architecture **100** adhere to the structure of the example class file **200** described in this section. However, in a practical environment, the structure of the class file **200** will be dependent on the implementation of the virtual machine **104**. Further, one or more features discussed herein may modify the structure of the class file **200** to, for example, add additional structure types. Therefore, the exact structure of the class file **200** is not critical to the techniques described herein. For the purposes of Section 2.1, “the class” or “the present class” refers to the class represented by the class file **200**.

(39) In FIG. 2, the class file **200** includes a constant table **201**, field structures **208**, class metadata **207**, and method structures **209**. In an embodiment, the constant table **201** is a data structure that,

among other functions, acts as a symbol table for the class. For example, the constant table **201** may store data related to the various identifiers used in the source code files **101** such as type, scope, contents, and/or location. The constant table **201** has entries for value structures **202** (representing constant values of type int, long, double, float, byte, string, and so forth), class information structures **203**, name and type information structures **204**, field reference structures **205**, and method reference structures **206** derived from the source code files **101** by the compiler **102**. In an embodiment, the constant table **201** is implemented as an array that maps an index *i* to structure *j*. However, the exact implementation of the constant table **201** is not critical.

(40) In some embodiments, the entries of the constant table **201** include structures that index other constant table **201** entries. For example, an entry for one of the value structures **202** representing a string may hold a tag identifying its “type” as string and an index to one or more other value structures **202** of the constant table **201** storing char, byte or int values representing the ASCII characters of the string.

(41) In an embodiment, field reference structures **205** of the constant table **201** hold an index into the constant table **201** to one of the class information structures **203** representing the class defining the field and an index into the constant table **201** to one of the name and type information structures **204** that provides the name and descriptor of the field. Method reference structures **206** of the constant table **201** hold an index into the constant table **201** to one of the class information structures **203** representing the class defining the method and an index into the constant table **201** to one of the name and type information structures **204** that provides the name and descriptor for the method. The class information structures **203** hold an index into the constant table **201** to one of the value structures **202** holding the name of the associated class.

(42) The name and type information structures **204** hold an index into the constant table **201** to one of the value structures **202** storing the name of the field/method and an index into the constant table **201** to one of the value structures **202** storing the descriptor.

(43) In an embodiment, class metadata **207** includes metadata for the class, such as version number(s), number of entries in the constant pool, number of fields, number of methods, access flags (whether the class is public, private, final, abstract, etc.), an index to one of the class information structures **203** of the constant table **201** that identifies the present class, an index to one of the class information structures **203** of the constant table **201** that identifies the superclass (if any), and so forth.

(44) In an embodiment, the field structures **208** represent a set of structures that identifies the various fields of the class. The field structures **208** store, for each field of the class, accessor flags for the field (whether the field is static, public, private, final, etc.), an index into the constant table **201** to one of the value structures **202** that holds the name of the field, and an index into the constant table **201** to one of the value structures **202** that holds a descriptor of the field.

(45) In an embodiment, the method structures **209** represent a set of structures that identifies the various methods of the class. The method structures **209** store, for each method of the class, accessor flags for the method (e.g. whether the method is static, public, private, synchronized, etc.), an index into the constant table **201** to one of the value structures **202** that holds the name of the method, an index into the constant table **201** to one of the value structures **202** that holds the descriptor of the method, and the virtual machine instructions that correspond to the body of the method as defined in the source code files **101**.

(46) In an embodiment, a descriptor represents a type of a field or method. For example, the descriptor may be implemented as a string adhering to a particular syntax. While the exact syntax is not critical, a few examples are described below.

(47) In an example where the descriptor represents a type of the field, the descriptor identifies the type of data held by the field. In an embodiment, a field can hold a basic type, an object, or an array. When a field holds a basic type, the descriptor is a string that identifies the basic type (e.g., “B”=byte, “C”=char, “D”=double, “F”=float, “I”=int, “J”=long int, etc.). When a field holds an

object, the descriptor is a string that identifies the class name of the object (e.g., “L ClassName”). “L” in this case indicates a reference, thus “L ClassName” represents a reference to an object of class ClassName. When the field is an array, the descriptor identifies the type held by the array. For example, “[B” indicates an array of bytes, with “[” indicating an array and “B” indicating that the array holds the basic type of byte. However, since arrays can be nested, the descriptor for an array may also indicate the nesting. For example, “[L ClassName” indicates an array where each index holds an array that holds objects of class ClassName. In some embodiments, the ClassName is fully qualified and includes the simple name of the class, as well as the pathname of the class. For example, the ClassName may indicate where the file is stored in the package, library, or file system hosting the class file **200**.

(48) In the case of a method, the descriptor identifies the parameters of the method and the return type of the method. For example, a method descriptor may follow the general form “({ParameterDescriptor}) ReturnDescriptor”, where the {ParameterDescriptor} is a list of field descriptors representing the parameters and the ReturnDescriptor is a field descriptor identifying the return type. For instance, the string “V” may be used to represent the void return type. Thus, a method defined in the source code files **101** as “Object m (int I, double d, Thread t) { . . . }” matches the descriptor “(I D L Thread) L Object”.

(49) In an embodiment, the virtual machine instructions held in the method structures **209** include operations that reference entries of the constant table **201**. Using Java as an example, consider the following class:

(50) TABLE-US-00001 class A { int add12and13() { return B.addTwo(12, 13); } }

(51) In the above example, the Java method add12and13 is defined in class A, takes no parameters, and returns an integer. The body of method add12 and13 calls static method addTwo of class B. The static method addTwo of class B takes the constant integer values **12** and **13** as parameters, and returns the result. Thus, in the constant table **201**, the compiler **102** includes, among other entries, a method reference structure that corresponds to the call to the method B.addTwo. In Java, a call to a method compiles down to an invoke command in the bytecode of the JVM (in this case invokestatic as addTwo is a static method of class B). The invoke command is provided an index into the constant table **201** corresponding to the method reference structure that identifies the class defining addTwo “B”, the name of addTwo “addTwo”, and the descriptor of addTwo “(I I)I”. For example, assuming the aforementioned method reference is stored at index 4, the bytecode instruction may appear as “invokestatic #4”.

(52) Since the constant table **201** refers to classes, methods, and fields symbolically with structures carrying identifying information, rather than direct references to a memory location, the entries of the constant table **201** are referred to as “symbolic references”. One reason that symbolic references are utilized for the class files **103** is because, in some embodiments, the compiler **102** is unaware of how and where the classes will be stored once loaded into the runtime environment **113**. As will be described in Section 2.3, eventually the run-time representations of the symbolic references are resolved into actual memory addresses by the virtual machine **104** after the referenced classes (and associated structures) have been loaded into the runtime environment and allocated concrete memory locations.

3.2 Example Virtual Machine Architecture

(53) FIG. 3 illustrates an example virtual machine memory layout **300** in block diagram form according to an embodiment. To provide clear examples, the remaining discussion will assume that the virtual machine **104** adheres to the virtual machine memory layout **300** depicted in FIG. 3. In addition, although components of the virtual machine memory layout **300** may be referred to as memory “areas”, there is no requirement that the memory areas be contiguous.

(54) In the example illustrated by FIG. 3, the virtual machine memory layout **300** is divided into a shared area **301** and a thread area **307**. The shared area **301** represents an area in memory where structures shared among the various threads executing on the virtual machine **104** are stored. The

shared area **301** includes a heap **302** and a per-class area **303**. In an embodiment, the heap **302** represents the run-time data area where memory for class instances and arrays is allocated. In an embodiment, the per-class area **303** represents the memory area where the data pertaining to the individual classes are stored. In an embodiment, the per-class area **303** includes, for each loaded class, a run-time constant pool **304** representing data from the constant table **201** of the class, field, and method data **306** (for example, to hold the static fields of the class), and the method code **305** representing the virtual machine instructions for methods of the class.

(55) The thread area **307** represents a memory area where structures specific to individual threads are stored. In FIG. 3, the thread area **307** includes thread structures **308** and thread structures **311**, representing the per-thread structures utilized by different threads. To provide clear examples, the thread area **307** depicted in FIG. 3 assumes two threads are executing on the virtual machine **104**. However, in a practical environment, the virtual machine **104** may execute any arbitrary number of threads, with the number of thread structures scaled accordingly.

(56) In an embodiment, thread structures **308** includes program counter **309** and virtual machine stack **310**. Similarly, thread structures **311** includes program counter **312** and virtual machine stack **313**. In an embodiment, program counter **309** and program counter **312** store the current address of the virtual machine instruction being executed by their respective threads.

(57) Thus, as a thread steps through the instructions, the program counters are updated to maintain an index to the current instruction. In an embodiment, virtual machine stack **310** and virtual machine stack **313** each store frames for their respective threads that hold local variables and partial results, and is also used for method invocation and return.

(58) In an embodiment, a frame is a data structure used to store data and partial results, return values for methods, and perform dynamic linking. A new frame is created each time a method is invoked. A frame is destroyed when the method that caused the frame to be generated completes. Thus, when a thread performs a method invocation, the virtual machine **104** generates a new frame and pushes that frame onto the virtual machine stack associated with the thread.

(59) When the method invocation completes, the virtual machine **104** passes back the result of the method invocation to the previous frame and pops the current frame off of the stack. In an embodiment, for a given thread, one frame is active at any point. This active frame is referred to as the current frame, the method that caused generation of the current frame is referred to as the current method, and the class that the current method belongs is referred to as the current class.

(60) Referring further to FIG. 3, in one example, the shared area may include the heap **302** and an off-heap memory **314**. One or more memory regions may be located within the heap **302**. Each memory region located in the heap **302** may include one or more shared memory segments that are accessible by a set of accessing threads. The off-heap memory **314** may include one or more memory regions located outside the heap **302**. Each memory region located in the off-heap memory **314** may include one or more shared memory segments that are accessible by a set of accessing threads. In one example, the off-heap memory **314** and/or a memory region located in the off-heap memory **314** may be managed by an application or by a provisioning thread, for example, instead of the JVM. In one example, the off-heap memory **314** and/or a memory region located in the off-heap memory **314** may be utilized for non-managed data structures, such as large buffers, caches, or custom data models. Additionally, or alternatively, the off-heap memory **314** and/or a memory region located in the off-heap memory **314** may be utilized for data structures that exceed the capacity of the heap **302** and/or that require more efficient memory handling. Additionally, or alternatively, the off-heap memory **314** and/or a memory region located in the off-heap memory **314** may be utilized to provide interaction with native libraries or to provide direct memory access.

(61) FIG. 4 illustrates an example frame **400** in block diagram form according to an embodiment. To provide clear examples, the remaining discussion will assume that frames of virtual machine stack **310** and virtual machine stack **313** adhere to the structure of frame **400**.

(62) In an embodiment, frame **400** includes local variables **401**, operand stack **402**, and run-time

constant pool reference table **403**. In an embodiment, the local variables **401** are represented as an array of variables that each hold a value, for example, Boolean, byte, char, short, int, float, or reference. Further, some value types, such as longs or doubles, may be represented by more than one entry in the array. The local variables **401** are used to pass parameters on method invocations and store partial results. For example, when generating the frame **400** in response to invoking a method, the parameters may be stored in predefined positions within the local variables **401**, such as indexes 1-N corresponding to the first to Nth parameters in the invocation.

(63) In an embodiment, the operand stack **402** is empty by default when the frame **400** is created by the virtual machine **104**. The virtual machine **104** then supplies instructions from the method code **305** of the current method to load constants or values from the local variables **401** onto the operand stack **402**. Other instructions take operands from the operand stack **402**, operate on them, and push the result back onto the operand stack **402**. Furthermore, the operand stack **402** is used to prepare parameters to be passed to methods and to receive method results. For example, the parameters of the method being invoked could be pushed onto the operand stack **402** prior to issuing the invocation to the method. The virtual machine **104** then generates a new frame for the method invocation where the operands on the operand stack **402** of the previous frame are popped and loaded into the local variables **401** of the new frame. When the invoked method terminates, the new frame is popped from the virtual machine stack and the return value is pushed onto the operand stack **402** of the previous frame.

(64) In an embodiment, the run-time constant pool reference table **403** contains a reference to the run-time constant pool **304** of the current class. The run-time constant pool reference table **403** is used to support resolution. Resolution is the process whereby symbolic references in the constant pool **304** are translated into concrete memory addresses, loading classes as necessary to resolve as-yet-undefined symbols and translating variable accesses into appropriate offsets into storage structures associated with the run-time location of these variables.

3.3 Loading, Linking, and Initializing

(65) In an embodiment, the virtual machine **104** dynamically loads, links, and initializes classes. Loading is the process of finding a class with a particular name and creating a representation from the associated class file **200** of that class within the memory of the runtime environment **113**. For example, creating the run-time constant pool **304**, method code **305**, and field and method data **306** for the class within the per-class area **303** of the virtual machine memory layout **300**. Linking is the process of taking the in-memory representation of the class and combining it with the run-time state of the virtual machine **104** so that the methods of the class can be executed. Initialization is the process of executing the class constructors to set the starting state of the field and method data **306** of the class and/or create class instances on the heap **302** for the initialized class.

(66) The following are examples of loading, linking, and initializing techniques that may be implemented by the virtual machine **104**. However, in some embodiments the steps may be interleaved, such that an initial class is loaded, then during linking a second class is loaded to resolve a symbolic reference found in the first class. The loading of the second class, in turn, causes a third class to be loaded, and so forth. Thus, progress through the stages of loading, linking, and initializing can differ from class to class. Further, some embodiments may delay (perform “lazily”) one or more functions of the loading, linking, and initializing process until the class is actually required. For example, resolution of a method reference may be delayed until a virtual machine instruction invoking the method is executed. Thus, the exact timing of when the steps are performed for each class can vary between implementations.

(67) To begin the loading process, the virtual machine **104** starts up by invoking the class loader **107**, and the class loader **107** loads an initial class. The technique whereby the initial class is specified will vary from embodiment to embodiment. For example, one technique may have the virtual machine **104** accept a command line argument on startup that specifies the initial class.

(68) To load a class, the class loader **107** parses the class file **200** corresponding to the class and

determines whether the class file **200** is well-formed (meets the syntactic expectations of the virtual machine **104**). If not, the class loader **107** generates an error. For example, in Java the error might be generated in the form of an exception that is thrown to an exception handler for processing. Otherwise, the class loader **107** generates the in-memory representation of the class by allocating the run-time constant pool **304**, method code **305**, and field and method data **306** for the class within the per-class area **303**.

(69) In some embodiments, when the class loader **107** loads a class, the class loader **107** also recursively loads the super-classes of the loaded class. For example, the virtual machine **104** may ensure that the super-classes of a particular class are loaded, linked, and/or initialized before proceeding with the loading, linking, and initializing process for the particular class.

(70) During linking, the virtual machine **104** verifies the class, prepares the class, and performs resolution of the symbolic references defined in the run-time constant pool **304** of the class.

(71) To verify the class, the virtual machine **104** checks whether the in-memory representation of the class is structurally correct. For example, the virtual machine **104** may check that each class except the generic class Object has a superclass, check that final classes have no sub-classes and final methods are not overridden, check whether constant pool entries are consistent with one another, check whether the current class has correct access permissions for classes/fields/structures referenced in the constant pool **304**, check that the virtual machine **104** code of methods will not cause unexpected behavior (e.g. making sure a jump instruction does not send the virtual machine **104** beyond the end of the method), and so forth. The exact checks performed during verification are dependent on the implementation of the virtual machine **104**. In some cases, verification may cause additional classes to be loaded, but does not necessarily require those classes to also be linked before proceeding. For example, assume Class A contains a reference to a static field of Class B. During verification, the virtual machine **104** may check Class B to ensure that the referenced static field actually exists, and this might cause loading of Class B, but not necessarily the linking or initializing of Class B. However, in some embodiments, certain verification checks can be delayed until a later phase, such as being checked during resolution of the symbolic references. For example, some embodiments may delay checking the access permissions for symbolic references until those references are being resolved.

(72) To prepare a class, the virtual machine **104** initializes static fields located within the field and method data **306** for the class to default values. In some cases, setting the static fields to default values may not be the same as running a constructor for the class. For example, the verification process may zero out or set the static fields to values that the constructor would expect those fields to have during initialization.

(73) During resolution, the virtual machine **104** dynamically determines concrete memory address from the symbolic references included in the run-time constant pool **304** of the class. To resolve the symbolic references, the virtual machine **104** utilizes the class loader **107** to load the class identified in the symbolic reference (if not already loaded). Once loaded, the virtual machine **104** has knowledge of the memory location within the per-class area **303** of the referenced class and its fields/methods. The virtual machine **104** then replaces the symbolic references with a reference to the concrete memory location of the referenced class, field, or method. In an embodiment, the virtual machine **104** caches resolutions to be reused in case the same class/name/descriptor is encountered when the virtual machine **104** processes another class. For example, in some cases, class A and class B may invoke the same method of class C. Thus, when resolution is performed for class A, that result can be cached and reused during resolution of the same symbolic reference in class B to reduce overhead.

(74) In some embodiments, the step of resolving the symbolic references during linking is optional. For example, an embodiment may perform the symbolic resolution in a “lazy” fashion, delaying the step of resolution until a virtual machine instruction that requires the referenced class/method/field is executed.

(75) During initialization, the virtual machine **104** executes the constructor of the class to set the starting state of that class. For example, initialization may initialize the field and method data **306** for the class and generate/initialize any class instances on the heap **302** created by the constructor. For example, the class file **200** for a class may specify that a particular method is a constructor that is used for setting up the starting state. Thus, during initialization, the virtual machine **104** executes the instructions of that constructor.

(76) In some embodiments, the virtual machine **104** performs resolution on field and method references by initially checking whether the field/method is defined in the referenced class. Otherwise, the virtual machine **104** recursively searches through the super-classes of the referenced class for the referenced field/method until the field/method is located or until the top-level superclass is reached. If the top-level super class is reached, an error may be generated.

4. Example Operations

(77) Referring now to FIGS. **5A-5C**, operations **500** pertaining to selecting GC processes are further described. The operations **500** described with reference to FIG. **5A** may include selecting a GC process from a group of GC processes to be executed upon a particular thread or upon a set of threads. The operations **500** described with reference to FIG. **5B** may include disqualifying a particular thread from eligibility for execution of a sweeping TLGC process based at least in part on the thread having stored a pointer in a shared memory region that includes a reference to a shared object located in a private memory region. The operations **500** described with reference to FIG. **5C** may include disqualifying a particular thread from eligibility for execution of a sweeping TLGC process based at least in part on the thread having loaded a pointer that includes a reference to a shared object located in a private memory region. The operations **500** described with reference to FIGS. **5A-5C** may be executed with respect to any type of thread, including a virtual thread, a carrier thread, a platform thread, and/or an operating system thread that is managed within a runtime environment by a platform thread.

(78) One or more operations **500** described with reference to FIGS. **5A-5C** may be executed using one or more components of the computing architecture described with reference to FIGS. **1-4**. One or more operations **500** described with reference to FIGS. **5A-5C** may be modified, combined, rearranged, or omitted. Accordingly, the particular sequence of operations **500** described with reference to FIGS. **5A-5C** should not be construed as limiting the scope of one or more embodiments.

4.1 Selecting a Garbage Collection Processes

(79) Referring to FIG. **5A**, operations pertaining to selecting GC processes are further described. As shown in FIG. **5A**, the operations **500** may include, at block **502**, executing a thread. In one example, executing the thread may include storing a set of objects in a private memory region. Additionally, or alternatively, executing the thread may include storing a set of pointers in the private memory region. The pointers may include a reference to a location of a particular object in the private memory region. The private memory region may include at least a portion of a thread area as described with reference to FIG. **3**. In one example, the private memory region may include a set of memory blocks allocated for the thread. Additionally, or alternatively, the private memory region may include a set of thread structures and/or a virtual machine stack associated with the thread. In one example, executing the thread may include storing a set of objects in a shared memory region. Additionally, or alternatively, executing the thread may include storing a set of pointers in the shared memory region. The shared memory region may include at least a portion of the shared area as described with reference to FIG. **3**. In one example, the shared memory region may include at least a portion of a heap corresponding to the shared area.

(80) At block **504**, the operations **500** may include detecting termination of the thread. The termination of the thread may be detected by a thread status checking algorithm that queries the status of the thread. In one example, in Java, the “Thread.isAlive()” operator may be utilized to check whether a thread is still running. Additionally, or alternatively, a thread can register a

callback or signal that gets triggered upon termination. Additionally, or alternatively, a thread variable may indicate whether a thread is active or terminated. An event handler may monitor the thread variable for one or more threads to determine threads are active and/or to determine when a thread is terminated. Additionally, or alternatively, an exception handling mechanism may detect thread termination events.

(81) At block **506**, the operations **500** may include determining whether or not the set of objects is exclusive of any shared objects that are accessible by one or more additional threads. The determination as to whether or not the set of objects is exclusive of any shared objects that are accessible by one or more additional threads may be performed at block **506** to select a GC process from a group of GC processes. The group of GC processes may include a sweeping TLGC process and a selective GC process. When the set of objects is exclusive of any shared objects that are accessible by one or more additional threads, the operations may proceed to block **508** and a sweeping TLGC process may be selected. When the set of objects includes at least one shared object that is accessible by one or more additional threads, the operations may proceed to block **510**, and a selective GC process may be selected.

(82) The determination as to whether or not the set of objects is exclusive of any shared objects that are accessible by one or more additional threads may be performed at least by determining whether respective objects are private or shared. In one example, the determination may be performed when the respective objects are stored and/or when the respective objects are loaded. Additionally, or alternatively, the determination as to whether or not the set of objects is exclusive of any shared objects that are accessible by one or more additional threads may be performed at least by determining whether pointers that respectively point to an object are shared pointers or private pointers. In one example, the determination may be performed when the respective pointers are stored and/or when the respective pointers are loaded.

(83) In one example, executing a thread at block **502** may include executing a plurality of store operations. The plurality of store operations may include storing a plurality of objects in a private memory region. Additionally, or alternatively, the plurality of store operations may include storing a plurality of pointers. At block **506**, the determination as to whether or not the set of objects is exclusive of any shared objects that are accessible by one or more additional threads may be performed at least in part at a store barrier when storing a respective object and/or when storing a respective pointer. Each store operation corresponding to an object may be interrogated at the store barrier by executing an object interrogation operation to determine whether the store operation includes storing, in a private memory region, a shared object that is accessible by one or more additional threads. Additionally, or alternatively, each store operation corresponding to a pointer may be interrogated at the store barrier by executing a pointer interrogation operation to determine whether the store operation includes storing, in a shared memory region, a pointer that includes a reference to a location of a shared object in a private memory region. When a thread executes a store operation that results in a shared object that is accessible by one or more additional threads, the thread may be disqualified from eligibility for execution of the sweeping TLGC process. Additionally, or alternatively, a thread may be eligible for the sweeping TLGC process when the objects resulting from the store operations executed by the thread are exclusive of any shared object that is accessible by one or more additional threads.

(84) In one example, executing a thread at block **502** may include executing a plurality of load operations. The plurality of load operations may include loading a plurality of pointers. Additionally, or alternatively, the plurality of load operations may include loading a plurality of objects from a private memory region. At block **506**, the determination as to whether or not the set of objects is exclusive of any shared objects that are accessible by one or more additional threads may be performed at least in part at a load barrier when loading a respective pointer and/or when loading a respective object. Each load operation corresponding to a pointer may be interrogated at the load barrier by executing a pointer interrogation operation to determine whether the load

operation includes loading, from a shared memory region, a pointer that includes a reference to a location of a shared object in a private memory region. Additionally, or alternatively, each load operation corresponding to an object may be interrogated at the load barrier by executing an object interrogation operation to determine whether or not the load operation includes loading, from a private memory region, a shared object that is accessible by one or more additional threads. When a thread executes a load operation that results in a shared object that is accessible by one or more additional threads, the thread may be disqualified from eligibility for execution of the sweeping TLGC process. Additionally, or alternatively, a thread may be eligible for the sweeping TLGC process when the objects resulting from the load operations executed by the thread are exclusive of any shared object that is accessible by one or more additional threads.

(85) At block **508**, the operations **500** may include executing the sweeping TLGC process. The sweeping TLGC process may be executed upon or after detecting termination of the thread at block **504**. The sweeping TLGC process may include reclaiming the private memory region. In one example, reclaiming the private memory region may include releasing, from the private memory region, a set of memory blocks allocated for the set of objects. Additionally, or alternatively, reclaiming the private memory region may include marking the set of memory blocks as available for reuse. Additionally, or alternatively, reclaiming the private memory region may include allocating the set of memory blocks to an available memory pool. In one example, the sweeping TLGC process may be executed asynchronously from termination of the thread. In one example, the sweeping TLGC process may be executed when a threshold level of memory blocks may be available to reclaim by executing the TLGC process. Additionally, or alternatively, the sweeping TLGC process may be postponed until the threshold level of memory blocks become available.

(86) At block **510**, the operations **500** may include executing the selective GC process. The selective GC process may include selectively reclaiming from the private memory region a subset of memory blocks allocated for a subset of private objects, of the set of objects, that are inaccessible from any thread. In one example, reclaiming the subset of memory blocks from the private memory region may include releasing the subset of memory blocks from the private memory region. Additionally, or alternatively, reclaiming the subset of memory blocks from the private memory region may include marking the subset of memory blocks as available for reuse. Additionally, or alternatively, reclaiming the subset of memory blocks from the private memory region may include allocating the subset of memory blocks to an available memory pool. The selective GC process may be executed upon or after detecting termination of the thread at block **504**. In one example, the selective GC process may be executed asynchronously from termination of the thread. In one example, the selective GC process may be executed when a threshold level of memory blocks may be available to reclaim by executing the selective GC process. Additionally, or alternatively, the selective GC process may be postponed until the threshold level of memory blocks become available.

(87) In one example, the operations **500** may include executing a first thread, including storing a first set of objects to a first private memory region. The operations **500** may also include detecting termination of the first thread, and executing a first GC process. In one example, the first GC process may include a sweeping TLGC process. The sweeping TLGC process may be selected for execution responsive to determining that the first set of objects is exclusive of any shared objects accessible by one or more additional threads. Execution of the sweeping TLGC process may include reclaiming the first private memory region. Additionally, or alternatively, the operations may include executing a second thread, including storing a second set of objects to a second private memory region. The operations **500** may also include detecting termination of the second thread and executing a second GC process. In one example, the second GC process may include a selective GC process. The selective GC process may be selected for execution responsive to determining that the second set of objects includes at least one shared object that is accessible by one or more additional threads. Execution of the selective GC process may include selectively

reclaiming from the second private memory region a subset of memory blocks allocated for a subset of private objects, of the second set of objects, that are inaccessible from any thread. In contrast with the selective GC process, the sweeping GC process may include reclaiming an entirety of the first private memory region without selectively reclaiming any subset of the first private memory region.

(88) In one example, executing the first thread may include storing a first set of pointers that respectively include a reference to the first private memory region. Execution of the sweeping TLGC process may be responsive to determining that the first set of pointers is exclusive of any pointers referencing a shared object that is accessible by one or more additional threads. Additionally, or alternatively, executing the first thread may include loading a first set of pointers. Execution of the sweeping TLGC process may be responsive to determining that the first set of pointers is exclusive of any pointer that includes a reference to a shared object stored in a private memory region.

(89) In one example, executing the second thread may include storing a second pointer that includes a reference to a shared object in the second private memory region. The selective GC process may be selected and executed responsive to determining that the second pointer includes the reference to the shared object in the second private memory region. Additionally, or alternatively, executing the second thread may include loading a third pointer that includes a reference to a shared object in a third private memory region. The selective GC process may be selected and executed responsive to loading the third pointer that includes the reference to the shared object in the third private memory region.

(90) The selective GC process may include a thread-local selective GC process or a global selective GC process. The thread-local selective GC process may include identifying and reclaiming memory allocated for objects in the second private memory region that are unreachable from any thread. The global selective GC process may include identifying and reclaiming from a plurality of private memory regions, including the second private memory region, memory allocated for objects that are unreachable from any thread.

4.2. Disqualifying Threads from Sweeping TLGC

(91) Referring to FIG. 5B, the operations **500** pertaining to disqualifying threads from sweeping TLGC are further described. As shown in FIG. 5B, the operations **500** may include, at block **520**, executing a first virtual thread. Executing the first virtual thread may include storing a first set of one or more objects in a first private memory region and storing a pointer in a shared memory region. The first private memory region may include at least a portion of a thread area as described with reference to FIG. 3. The first private memory region may include a set of memory blocks allocated for the first virtual thread. Additionally, or alternatively, the first private memory region may include a set of thread structures and/or a virtual machine stack associated with the first virtual thread. The shared memory region may include at least a portion of the shared area as described with reference to FIG. 3. In one example, the shared memory region may include at least a portion of a heap corresponding to the shared area.

(92) At block **522**, the operations **500** may include determining whether the pointer includes a reference to a location of a shared object in the first private memory region. Additionally, or alternatively, the operations **500** at block **522** may include determining whether the first set of one or more objects stored in the first private memory region includes a shared object. If the reference of pointer points to a shared object in the first private memory region, then the shared object may be accessible by other virtual threads via the pointer stored in the shared memory region. When the pointer stored in the shared memory region at block **520** includes a reference to a location of a shared object in the first private memory region, the shared object may be one of the first set of one or more objects stored in the first private memory region at block **520**. In one example, the pointer may be accessed by a second virtual thread as described with reference to FIG. 5C. When the pointer includes a reference to a location of the shared object in the first private memory region, the

operations may proceed to block **524**. When the pointer does not include a reference to a location of the shared object in the first private memory region, the operations may proceed to block **532**.

(93) At block **524**, the operations **500** may include designating the first virtual thread as disqualified from eligibility for execution of a sweeping TLGC process. In one example, the system may designate the first virtual thread as disqualified from eligibility for execution of the sweeping TLGC process based on a thread-specific flag. The flag may be associated with the first virtual thread or a thread ID corresponding to the first virtual thread. Additionally, or alternatively, the system may allocate a data structure to the first virtual thread to track whether or not the first virtual thread is disqualified from eligibility for execution of the sweeping TLGC process. The system may update the data structure allocated to the first virtual thread to indicate that the first virtual thread is disqualified.

(94) At block **526**, the operations **500** may include augmenting the pointer to indicate that the pointer includes the reference to the location of the shared object in the first private memory region. In one example, the pointer may include a pointer attribute bitset, and augmenting the pointer may include augmenting the pointer attribute bitset to indicate that the pointer includes the reference to the location of the first shared object in the first private memory region.

(95) In one example, augmenting the pointer may be configured to trigger an error in response to an additional virtual thread loading the pointer. The error may trigger execution of an error-handling operation on the pointer that may be utilized to determine that the pointer has been augmented as described with reference to FIG. 5C. In one example, augmenting the pointer may include repositioning one or more bitsets of the pointer. Repositioning at least the portion of the metadata bitset may trigger an error in response to an additional virtual thread loading the pointer.

(96) In one example, the pointer may include a resource location bitset and a metadata bitset, and augmenting the pointer may include repositioning at least a portion of the metadata bitset relative to the resource location bitset. The resource location bitset may include a set of one or more resource location bits representing the location of the shared object. The metadata bitset may include a set of one or more metadata bits representing information pertaining to the pointer and/or the shared object. The repositioning of at least the portion of the metadata bitset relative to the resource location bitset may include at least one of the following: a bit-shifting operation, a bit-reversal operation, or a bit-swapping operation. The bit-shifting operation, the bit-reversal operation, and/or the bit-swapping operation may be executed upon the metadata bitset or the resource location bitset.

(97) In one example, the determination at block **522**, as to whether or not the pointer includes the reference to the location of the shared object in the first private memory region, may be performed at a store barrier. In one example, executing the first virtual thread at block **520** may include executing a store operation for storing the pointer in the shared memory region. Additionally, the operations at block **522** may include intercepting the store operation at a store barrier, triggering, at the store barrier, execution of a pointer interrogation operation, and determining that the pointer includes the reference to the location of the first shared object in the first private memory region at least by executing the pointer interrogation operation. Additionally, or alternatively, designating the first virtual thread as disqualified from eligibility for execution of a sweeping TLGC process may include executing a disqualification operation. The disqualification operation may be triggered and executed at the store barrier. The first virtual thread may be designated as disqualified from eligibility for the sweeping TLGC process at least by executing the disqualification operation. Additionally, or alternatively, augmenting the pointer at block **526** may include executing a pointer augmentation operation. The pointer augmentation operation may be triggered and executed at the store barrier. The pointer augmentation operation may be executed responsive to determining that the pointer includes the reference to the location of the first shared object in the first private memory region. The pointer may be augmented at least by executing the pointer augmentation operation.

(98) At block **528**, the operations **500** may include exiting the first virtual thread. The first virtual thread may exit at the end of its designated set of tasks. Additionally, or alternatively, the first virtual thread may be a part of a larger process, and the first virtual thread may exit when the larger process exits. Additionally, or alternatively, the first virtual thread may exit as a result of an explicit termination instruction, a signal, or an interrupt. Additionally, or alternatively, the first virtual thread may exit as a result of an exception, an error, or a timeout.

(99) At block **530**, the operations **500** may include executing a selective GC process with respect to the first virtual thread. The selective GC process may be executed upon the first virtual thread having exited, such as immediately after the first virtual thread exits or after a period of time has elapsed. The selective GC process may be triggered based at least in part on the first virtual thread having exited. Additionally, or alternatively, the system may initiate the selective GC process when or after the first virtual thread exits. In one example, the system may execute the selective GC process in response to determining that the first virtual thread is disqualified from eligibility for execution of a sweeping TLGC process. In one example, the system may check a thread-specific flag or a data structure associated with the first virtual thread to determine whether or not the first virtual thread has been disqualified from eligibility for execution of the sweeping TLGC process.

(100) The exiting of the first virtual thread may be detected by a thread status checking algorithm that queries the status of the first virtual thread. In one example, in Java, the “Thread.isAlive()” operator may be utilized to check whether the first virtual thread is still executing. Additionally, or alternatively, the first virtual thread can register a callback or signal that gets triggered when the first virtual thread exits. Additionally, or alternatively, a thread variable may indicate whether the first virtual thread is active or exited. An event handler may monitor the thread variable for one or more threads to determine threads that are active and/or to determine when the first virtual thread is exited. Additionally, or alternatively, an exception handling mechanism may detect thread exit events.

(101) In one example, the selective GC process may include a global selective GC process. The global selective GC process may be executed with respect to a plurality of virtual threads after exiting the plurality of virtual threads. In one example, the selective GC process may be executed with respect to the first virtual thread and one or more additional threads. The one or more additional threads may include the second virtual thread described with reference to FIG. 5C. The global selective GC process may include identifying and reclaiming from a plurality of memory regions corresponding to the plurality of virtual threads, memory allocated for objects that are unreachable from any thread. Additionally, or alternatively, the selective GC process may include a thread-local selective GC process. The thread-local selective GC process may be executed with respect to the first virtual thread. The thread-local selective GC process may include identifying and reclaiming memory allocated for objects in the first private memory region that are unreachable from any thread.

(102) At block **532**, the operations **500** may include exiting the first virtual thread. As described with reference to block **528**, the first virtual thread may exit upon the occurrence of one or more of the following: at the end of its designated set of tasks; when a larger process exits; as a result of an explicit termination instruction, a signal, or an interrupt; as a result of an exception, an error, or a timeout.

(103) At block **534**, the operations **500** may include executing a sweeping TLGC process with respect to the first virtual thread. The sweeping TLGC process may include reclaiming the first private memory region. The sweeping TLGC process may be executed upon the first virtual thread having exited, such as immediately after the first virtual thread exits or after a period of time has elapsed. The sweeping TLGC process may be triggered based at least in part on the first virtual thread having exited. Additionally, or alternatively, the system may initiate the sweeping TLGC process when or after the first virtual thread exits. In one example, the system may execute the sweeping TLGC process in response to determining that the first virtual thread is eligible for

execution of the sweeping TLGC process. In one example, the system may check a thread-specific flag or a data structure associated with the first virtual thread to determine whether or not the first virtual thread is eligible for execution of the sweeping TLGC process.

(104) As described with reference to block **530**, the exiting of the first virtual thread may be detected by one or more of the following: a thread status checking algorithm; a callback or signal that gets triggered when the first virtual thread exits; a thread variable that indicates whether the first virtual thread is active or exited; an event handler that keeps track of threads that are active and/or exited; or an exception handling mechanism that detects thread termination events.

(105) Referring to FIG. 5C, the operations **500** pertaining to disqualifying threads from sweeping TLGC are further described. As shown in FIG. 5B, the operations **500** may include, at block **540**, executing a second virtual thread. Executing the second virtual thread may include loading a pointer. The pointer loaded at block **540** may have been stored by a virtual thread other than the second virtual thread. In one example, the pointer loaded at block **540** may be the pointer stored when executing the first virtual thread, for example, at block **520**. Additionally, executing the second virtual thread may include storing one or more objects to a second private memory region. The second private memory region may be allocated for the second virtual thread. The second private memory region may be a different private memory region from the first private memory region described with reference to FIG. 5B. The second private memory region may include at least a portion of a thread area as described with reference to FIG. 3. The second private memory region may include a set of memory blocks allocated for the second virtual thread. Additionally, or alternatively, the second private memory region may include a set of thread structures and/or a virtual machine stack associated with the second virtual thread.

(106) At block **542**, the operations **500** may include determining whether the pointer loaded by the second virtual thread at block **540** has been augmented to indicate that the pointer includes a reference to a location of a shared object in a private memory region. In one example, the pointer loaded by the second virtual thread at block **540** may include a reference to a location in the first private memory region allocated for the first virtual thread described with reference to FIG. 5B. Additionally, or alternatively, the pointer may include a reference to a location in another private memory region allocated for another virtual thread. Additionally, or alternatively, the pointer loaded by the second virtual thread at block **540** may not include a reference to a location of a shared object in a private memory region. In one example, the pointer loaded by the second virtual thread at block **540** may include a reference to a location of a private object in the second private memory region or a reference to a location of a shared object in a shared memory region. When the pointer has been augmented to indicate that the pointer includes a reference to a location of a shared object in a private memory region, the operations may proceed to block **544**. When the pointer has not been augmented to indicate that the pointer includes a reference to a location of a shared object in a private memory region, the operations may proceed to block **550**.

(107) In one example, the pointer may have been augmented in a manner that is configured to trigger an error in response to a virtual thread loading the pointer. In one example, the operations **500** may include determining the error responsive to the second virtual thread loading the pointer. Additionally, the operations **500** may include executing an error-handling operation on the pointer. The error-handling operation may be triggered in response to the error. In one example, the error-handling operation may include determining that the pointer has been augmented to indicate that the pointer includes a reference to a location of a shared object in a private memory region. For example, the pointer loaded by the second virtual thread may have been augmented to include a reference to the location of the first shared object in the first private memory region as described with reference to FIG. 5B.

(108) In one example, executing the error-handling operation may include resolving the error associated with the pointer. Resolving the error may include determining the location of the first shared object in the first private memory region. In one example, the operations **500** may include

further executing the second virtual thread. Further executing the second virtual thread may include loading the first shared object from the first private memory region in response to resolving the error and/or in response to determining the location of the first shared object in the first private memory region.

(109) In one example, the determination at block **542**, as to whether or not the pointer loaded by the second virtual thread has been augmented, may be performed at a load barrier. In one example, executing the second virtual thread at block **540** may include executing a load operation for loading the pointer from the shared memory region. In addition, the operations at block **542** may include intercepting the load operation at a load barrier, triggering, at the load barrier, execution of a pointer interrogation operation and determining that the pointer includes the reference to the location of the first shared object in the first private memory region at least by executing the pointer interrogation operation. Additionally, or alternatively, the operations at block **542** may include determining that the pointer has been augmented to indicate that the pointer includes the reference to the location of the first shared object in the first private memory region at least by executing the pointer interrogation operation. Additionally, or alternatively, designating the second virtual thread as disqualified from eligibility for execution of a sweeping TLGC process may include executing a disqualification operation. The disqualification operation may be triggered and executed at the load barrier. The second virtual thread may be designated as disqualified from eligibility for the sweeping TLGC process at least by executing the disqualification operation.

(110) At block **544**, the operations **500** may include designating the second virtual thread as disqualified from eligibility for execution of the sweeping TLGC process. In one example, the system may designate the second virtual thread as disqualified from eligibility for execution of the sweeping TLGC process based on a thread-specific flag. The flag may be associated with the second virtual thread or a thread ID corresponding to the second virtual thread. Additionally, or alternatively, the system may allocate a data structure to the second virtual thread to track whether the second virtual thread is disqualified from eligibility for execution of the sweeping TLGC process. The system may update the data structure allocated to the second virtual thread to indicate that the second virtual thread is disqualified.

(111) At block **546**, the operations **500** may include exiting the second virtual thread. The second virtual thread may exit at the end of its designated set of tasks. Additionally, or alternatively, the second virtual thread may be a part of a larger process and the second virtual thread may exit when the larger process exits. Additionally, or alternatively, the second virtual thread may exit as a result of an explicit termination instruction, a signal, or an interrupt. Additionally, or alternatively, the second virtual thread may exit as a result of an exception, an error, or a timeout.

(112) At block **548**, the operations **500** may include executing a selective GC process with respect to the second virtual thread. The selective GC process may be executed upon the second virtual thread having exited, such as immediately after the second virtual thread exits or after a period of time has elapsed. The selective GC process may be triggered based at least in part on the second virtual thread having exited. Additionally, or alternatively, the system may initiate the selective GC process when or after the second virtual thread exits. In one example, the system may execute the selective GC process in response to determining that the second virtual thread is disqualified from eligibility for execution of a sweeping TLGC process. In one example, the system may check a thread-specific flag or a data structure associated with the second virtual thread to determine whether the second virtual thread has been disqualified from eligibility for execution of the sweeping TLGC process.

(113) The exiting of the second virtual thread may be detected by a thread status checking algorithm that queries the status of the second virtual thread. In one example, in Java, the “Thread.isAlive()” operator may be utilized to check whether the second virtual thread is still executing. Additionally, or alternatively, the second virtual thread can register a callback or signal that gets triggered when the second virtual thread exits. Additionally, or alternatively, a thread

variable may indicate whether the second virtual thread is active or exited. An event handler may monitor the thread variable for one or more threads to determine threads that are active and/or to determine when the second virtual thread is exited. Additionally, or alternatively, an exception handling mechanism may detect thread exit events.

(114) In one example, the selective GC process may include a global selective GC process. The global selective GC process may be executed with respect to a plurality of virtual threads after exiting the plurality of virtual threads. In one example, the selective GC process may be executed with respect to the first virtual thread and the second virtual thread. The global selective GC process may include identifying and reclaiming from a plurality of memory regions corresponding to the plurality of virtual threads, memory allocated for objects that are unreachable from any thread. Additionally, or alternatively, the selective GC process may include a thread-local selective GC process. The thread-local selective GC process may be executed with respect to the second virtual thread. The thread-local selective GC process may include identifying and reclaiming memory allocated for objects in the second private memory region that are unreachable from any thread.

(115) At block **550**, the operations **500** may include exiting the second virtual thread. As described with reference to block **546**, the second virtual thread may exit upon the occurrence of one or more of the following: at the end of its designated set of tasks; when a larger process exits; as a result of an explicit termination instruction, a signal, or an interrupt; as a result of an exception, an error, or a timeout.

(116) At block **552**, the operations **500** may include executing a sweeping TLGC process with respect to the second virtual thread. The sweeping TLGC process may include reclaiming the second private memory region. The sweeping TLGC process may be executed upon the second virtual thread having exited, such as immediately after the second virtual thread exits or after a period of time has elapsed. The sweeping TLGC process may be triggered based at least in part on the second virtual thread having exited. Additionally, or alternatively, the system may initiate the sweeping TLGC process when or after the second virtual thread exits. In one example, the system may execute the sweeping TLGC process in response to determining that the second virtual thread is eligible for execution of the sweeping TLGC process. In one example, the system may check a thread-specific flag or a data structure associated with the second virtual thread to determine whether the second virtual thread is eligible for execution of the sweeping TLGC process.

(117) As described with reference to block **548**, the exiting of the second virtual thread may be detected by one or more of the following: a thread status checking algorithm; a callback or signal that gets triggered when the second virtual thread exits; a thread variable that indicates whether the second virtual thread is active or exited; an event handler that keeps track of threads that are active and/or exited; or an exception handling mechanism that detects thread termination events.

5. Hardware Overview

(118) According to one embodiment, the techniques described herein are implemented by one or more special-purpose computing devices. The special-purpose computing devices may be hard-wired to perform the techniques, or may include digital electronic devices such as one or more application-specific integrated circuits (ASICs), field programmable gate arrays (FPGAs), or network processing units (NPU) that are persistently programmed to perform the techniques, or may include one or more general purpose hardware processors programmed to perform the techniques pursuant to program instructions in firmware, memory, other storage, or a combination. Such special-purpose computing devices may also combine custom hard-wired logic, ASICs, FPGAs, or NPUs with custom programming to accomplish the techniques. The special-purpose computing devices may be desktop computer systems, portable computer systems, handheld devices, networking devices or any other device that incorporates hard-wired and/or program logic to implement the techniques.

(119) For example, FIG. **6** is a block diagram that illustrates a computer system **600** that may be

utilized to implement at least one embodiment of the present disclosure. Computer system **600** may include a bus **602** or other communication mechanism for communicating information, and a hardware processor **604** coupled with bus **602** for processing information. Hardware processor **604** may be, for example, a general-purpose microprocessor.

(120) Computer system **600** also may include a main memory **606**, such as a random-access memory (RAM) or other dynamic storage device, coupled to bus **602** for storing information and instructions to be executed by processor **604**. Main memory **606** also may be used for storing temporary variables or other intermediate information during execution of instructions to be executed by processor **604**. Such instructions, when stored in non-transitory storage media accessible to processor **604**, render computer system **600** into a special-purpose machine that is customized to perform the operations specified in the instructions.

(121) Computer system **600** may further include a read only memory (ROM) **608** or other static storage device coupled to bus **602** for storing static information and instructions for processor **604**. A storage device **610**, such as a magnetic disk or optical disk, is provided and coupled to bus **602** for storing information and instructions.

(122) Computer system **600** may be coupled via bus **602** to a display **612**, such as a cathode ray tube (CRT), for displaying information to a computer user. An input device **614**, including alphanumeric and other keys, is coupled to bus **602** for communicating information and command selections to processor **604**. Another type of user input device is cursor control **616**, such as a mouse, a trackball, or cursor direction keys for communicating direction information and command selections to processor **604** and for controlling cursor movement on display **612**. This input device typically has two degrees of freedom in two axes, a first axis (e.g., x) and a second axis (e.g., y), that allows the device to specify positions in a plane.

(123) Computer system **600** may implement the techniques described herein using customized hard-wired logic, one or more ASICs or FPGAs, firmware and/or program logic that, in combination with the computer system, causes or programs computer system **600** to be a special-purpose machine. According to one embodiment, the techniques herein are performed by computer system **600** in response to processor **604** executing one or more sequences of one or more instructions contained in main memory **606**. Such instructions may be read into main memory **606** from another storage medium, such as storage device **610**. Execution of the sequences of instructions contained in main memory **606** causes processor **604** to perform the process steps described herein. In alternative embodiments, hard-wired circuitry may be used in place of or in combination with software instructions.

(124) The term “storage media” as used herein refers to any non-transitory media that store data and/or instructions that cause a machine to operate in a specific fashion. Such storage media may comprise non-volatile media and/or volatile media. Non-volatile media may include, for example, optical or magnetic disks, such as storage device **610**. Volatile media may include dynamic memory, such as main memory **606**. Common forms of storage media include, for example, a floppy disk, a flexible disk, hard disk, solid state drive, magnetic tape, or any other magnetic data storage medium, a CD-ROM, any other optical data storage medium, any physical medium with patterns of holes, a RAM, a PROM, and EPROM, a FLASH-EPROM, NVRAM, any other memory chip or cartridge, content-addressable memory (CAM), and ternary content-addressable memory (TCAM).

(125) Storage media is distinct from but may be used in conjunction with transmission media. Transmission media participates in transferring information between storage media. For example, transmission media may include coaxial cables, copper wire and fiber optics, including the wires that comprise bus **602**. Transmission media can also take the form of acoustic or light waves, such as those generated during radio-wave and infra-red data communications.

(126) Various forms of media may be involved in carrying one or more sequences of one or more instructions to processor **604** for execution. For example, the instructions may initially be carried

on a magnetic disk or solid-state drive of a remote computer. The remote computer can load the instructions into its dynamic memory and send the instructions over a telephone line using a modem. A modem local to computer system **600** can receive the data on the telephone line and use an infra-red transmitter to convert the data to an infra-red signal. An infra-red detector can receive the data carried in the infra-red signal and appropriate circuitry can place the data on bus **602**. Bus **602** carries the data to main memory **606**. Processor **604** retrieves the data from main memory **606** and executes the instructions. The instructions received from main memory **606** may optionally be stored on storage device **610** either before or after execution by processor **604**.

(127) Computer system **600** also may include a communication interface **618** coupled to bus **602**. Communication interface **618** provides a two-way data communication coupling to a network link **620** that is connected to a local network **622**. For example, communication interface **618** may be an integrated services digital network (ISDN) card, cable modem, satellite modem, or a modem to provide a data communication connection to a corresponding type of telephone line. As another example, communication interface **618** may be a local area network (LAN) card to provide a data communication connection to a compatible LAN. Wireless links may also be implemented. In any such implementation, communication interface **618** sends and receives electrical, electromagnetic, or optical signals that carry digital data streams representing various types of information.

(128) Network link **620** typically provides data communication through one or more networks to other data devices. For example, network link **620** may provide a connection through local network **622** to a host computer **624** or to data equipment operated by an Internet Service Provider (ISP) **626**. ISP **626** in turn provides data communication services through the world-wide packet data communication network now commonly referred to as the “Internet” **628**. Local network **622** and Internet **628** both use electrical, electromagnetic, or optical signals that carry digital data streams. Example forms of transmission media include the signals through the various networks, the signals through network link **620**, and the signals through communication interface **618**.

(129) Computer system **600** can send messages and receive data, including program code, through the network(s), network link **620** and communication interface **618**. In the Internet example, a server **630** might transmit a requested code for an application program through Internet **628**, ISP **626**, local network **622** and communication interface **618**. The received code may be executed by processor **604** as it is received, and/or stored in storage device **610**, or other non-volatile storage for later execution.

6. Miscellaneous; Extensions

(130) Embodiments are directed to a system with one or more devices that include a hardware processor and that are configured to perform any of the operations described herein and/or recited in any of the claims below.

(131) In an embodiment, a non-transitory computer readable storage medium comprises instructions that, when executed by one or more hardware processors, causes performance of any of the operations described herein and/or recited in any of the claims.

(132) Any combination of the features and functionalities described herein may be used in accordance with one or more embodiments. In the foregoing specification, embodiments have been described with reference to numerous specific details that may vary from implementation to implementation. The specification and drawings are, accordingly, to be regarded in an illustrative rather than a restrictive sense. The sole and exclusive indicator of the scope of patent protection, and what is intended by the applicants to be the scope of patent protection, is the literal and equivalent scope of the set of claims that issue from this application, in the specific form that such claims issue, including any subsequent correction.

Claims

1. One or more non-transitory computer-readable media storing instructions that, when executed by one or more hardware processors, cause performance of operations comprising: executing a first virtual thread, wherein executing the first virtual thread comprises: storing a first shared object in a first private memory region, storing a pointer in a shared memory region, and determining that the pointer comprises a reference to a location of the first shared object in the first private memory region; responsive to determining that the pointer comprises the reference to the location of the first shared object in the first private memory region: designating the first virtual thread as disqualified from eligibility for execution of a sweeping thread-local garbage collection process; and augmenting the pointer to indicate that the pointer comprises the reference to the location of the first shared object in the first private memory region.
2. The one or more non-transitory computer-readable media of claim 1, wherein the operations further comprise: executing a second virtual thread, wherein executing the second virtual thread comprises: loading the pointer, and determining that the pointer has been augmented to indicate that the pointer comprises the reference to the location of the first shared object in the first private memory region; responsive to determining that the pointer has been augmented to indicate that the pointer comprises the reference to the location of the first shared object in the first private memory region: designating the second virtual thread as disqualified from eligibility for execution of the sweeping thread-local garbage collection process.
3. The one or more non-transitory computer-readable media of claim 2, wherein augmenting the pointer is configured to trigger an error in response to an additional virtual thread loading the pointer; and wherein the operations further comprise: determining the error responsive to the second virtual thread loading the pointer, and responsive to determining the error, executing an error-handling operation on the pointer, wherein executing the error-handling operation comprises: determining that the pointer has been augmented to indicate that the pointer comprises the reference to the location of the first shared object in the first private memory region.
4. The one or more non-transitory computer-readable media of claim 3, wherein executing the error-handling operation further comprises: resolving the error associated with the pointer, wherein resolving the error comprises determining the location of the first shared object in the first private memory region; wherein executing the second virtual thread further comprises: responsive to determining the location of the first shared object in the first private memory region, and loading the first shared object from the first private memory region.
5. The one or more non-transitory computer-readable media of claim 2, wherein the operations further comprise: exiting the first virtual thread; exiting the second virtual thread; and executing a global selective garbage collection process with respect to the first virtual thread and the second virtual thread, wherein the global selective garbage collection process comprises identifying and reclaiming memory allocated for objects that are unreachable from any thread.
6. The one or more non-transitory computer-readable media of claim 1, wherein the pointer comprises a resource location bitset and a metadata bitset, wherein the resource location bitset comprises a set of one or more resource location bits representing the location of the first shared object, wherein the metadata bitset comprises a set of one or more metadata bits representing information pertaining to at least one of: the pointer, or the first shared object, wherein augmenting the pointer comprises: repositioning at least a portion of the metadata bitset relative to the resource location bitset, wherein repositioning at least the portion of the metadata bitset is configured to trigger an error in response to an additional virtual thread loading the pointer.
7. The one or more non-transitory computer-readable media of claim 6, wherein repositioning at least the portion of the metadata bitset relative to the resource location bitset comprises at least one of: executing a bit-shifting operation upon the metadata bitset or the resource location bitset; executing a bit-reversal operation upon the metadata bitset or the resource location bitset; executing a bit-swapping operation upon the metadata bitset or the resource location bitset; or executing a bit-

rotation operation upon the metadata bitset or the resource location bitset.

8. The one or more non-transitory computer-readable media of claim 1, wherein determining that the pointer comprises the reference to the location of the first shared object in the first private memory region comprises: executing a store operation for storing the pointer in the shared memory region; intercepting the store operation at a store barrier; triggering, at the store barrier, execution of a pointer interrogation operation; determining that the pointer comprises the reference to the location of the first shared object in the first private memory region at least by executing the pointer interrogation operation.

9. The one or more non-transitory computer-readable media of claim 8, wherein augmenting the pointer comprises: responsive to determining that the pointer comprises the reference to the location of the first shared object in the first private memory region, triggering at the store barrier, execution of a pointer augmentation operation; augmenting the pointer, at least by executing the pointer augmentation operation, to indicate that the pointer comprises the reference to the location of the first shared object in the first private memory region.

10. The one or more non-transitory computer-readable media of claim 8, wherein designating the first virtual thread as disqualified from eligibility for the sweeping thread-local garbage collection process comprises: responsive to determining that the pointer comprises the reference to the location of the first shared object in the first private memory region, triggering at the store barrier, execution of a disqualification operation; designating the first virtual thread as disqualified from eligibility for the sweeping thread-local garbage collection process at least by executing the disqualification operation.

11. The one or more non-transitory computer-readable media of claim 1, wherein the pointer comprises a pointer attribute bitset, wherein augmenting the pointer comprises: augmenting the pointer attribute bitset to indicate that the pointer comprises the reference to the location of the first shared object in the first private memory region.

12. The one or more non-transitory computer-readable media of claim 1, wherein the operations further comprise: exiting the first virtual thread; and executing a thread-local selective garbage collection process with respect to the first virtual thread, wherein the thread-local selective garbage collection process comprises identifying and reclaiming memory allocated for objects in the first private memory region that are unreachable from any thread.

13. The one or more non-transitory computer-readable media of claim 1, wherein the operations further comprise: executing a second virtual thread, wherein executing the second virtual thread comprises: executing a plurality of store operations comprising storing a plurality of objects in a second private memory region allocated for the second virtual thread, wherein the plurality of store operations are exclusive of storing in the second private memory region, any shared objects accessible by one or more additional virtual threads; exiting the second virtual thread; and executing the sweeping thread-local garbage collection process with respect to the second virtual thread, wherein the sweeping thread-local garbage collection process comprises at least one of: releasing, from the second private memory region, a first set of memory blocks allocated for the plurality of objects; marking the first set of memory blocks as available for reuse; or allocating the first set of memory blocks to an available memory pool.

14. The one or more non-transitory computer-readable media of claim 1, wherein the operations further comprise: executing a second virtual thread, wherein executing the second virtual thread comprises: executing a plurality of load operations comprising loading a plurality of pointers respectively comprising a reference to a second private memory region, wherein the plurality of load operations is exclusive of loading any pointers from a shared memory region comprising a reference to a third private memory region; exiting the second virtual thread; and executing the sweeping thread-local garbage collection process with respect to the second virtual thread, wherein the sweeping thread-local garbage collection process further comprises at least one of: releasing, from the second private memory region, a second set of memory blocks allocated for the plurality

of pointers; marking the second set of memory blocks as available for reuse; or allocating the second set of memory blocks to an available memory pool.

15. A method, comprising: executing a first virtual thread, wherein executing the first virtual thread comprises: storing a first shared object in a first private memory region, storing a pointer in a shared memory region, and determining that the pointer comprises a reference to a location of the first shared object in the first private memory region; responsive to determining that the pointer comprises the reference to the location of the first shared object in the first private memory region: designating the first virtual thread as disqualified from eligibility for execution of a sweeping thread-local garbage collection process; and augmenting the pointer to indicate that the pointer comprises the reference to the location of the first shared object in the first private memory region; wherein the method is performed by at least one device including a hardware processor.

16. The method of claim 15, further comprising: executing a second virtual thread, wherein executing the second virtual thread comprises: loading the pointer, and determining that the pointer has been augmented to indicate that the pointer comprises the reference to the location of the first shared object in the first private memory region; responsive to determining that the pointer has been augmented to indicate that the pointer comprises the reference to the location of the first shared object in the first private memory region: designating the second virtual thread as disqualified from eligibility for execution of the sweeping thread-local garbage collection process.

17. The method of claim 16, wherein augmenting the pointer is configured to trigger an error in response to an additional virtual thread loading the pointer; and wherein the method further comprises: determining the error responsive to the second virtual thread loading the pointer, and responsive to determining the error, executing an error-handling operation on the pointer, wherein executing the error-handling operation comprises: determining that the pointer has been augmented to indicate that the pointer comprises the reference to the location of the first shared object in the first private memory region.

18. The method of claim 15, wherein the pointer comprises a resource location bitset and a metadata bitset, wherein the resource location bitset comprises a set of one or more resource location bits representing the location of the first shared object, wherein the metadata bitset comprises a set of one or more metadata bits representing information pertaining to at least one of: the pointer, or the first shared object, wherein augmenting the pointer comprises: repositioning at least a portion of the metadata bitset relative to the resource location bitset, wherein repositioning at least the portion of the metadata bitset is configured to trigger an error in response to an additional virtual thread loading the pointer.

19. The method of claim 15, wherein determining that the pointer comprises the reference to the location of the first shared object in the first private memory region comprises: executing a store operation for storing the pointer in the shared memory region; intercepting the store operation at a store barrier; triggering, at the store barrier, execution of a pointer interrogation operation; determining that the pointer comprises the reference to the location of the first shared object in the first private memory region at least by executing the pointer interrogation operation.

20. A system, comprising: at least one hardware processor; wherein the system is configured to execute operations, using the at least one hardware processor, the operations comprising: executing a first virtual thread, wherein executing the first virtual thread comprises: storing a first shared object in a first private memory region, storing a pointer in a shared memory region, and determining that the pointer comprises a reference to a location of the first shared object in the first private memory region; responsive to determining that the pointer comprises the reference to the location of the first shared object in the first private memory region: designating the first virtual thread as disqualified from eligibility for execution of a sweeping thread-local garbage collection process; and augmenting the pointer to indicate that the pointer comprises the reference to the location of the first shared object in the first private memory region.
